LOCALIZED APPROXIMATION VARIANT: DESCENDING THE ETALE SOLUTION FROM R_P TO AN ETALE R-ALGEBRA

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Editorial note. The source proof is complete; no omitted steps.

Allowed standard prerequisites (editorial).

- Approximation with etale targets. theorem-approximation-property-variant (carded L26821). Supplies the solution over an etale extension of R_p with trivial residue extension.
- Descent of etale maps along localizations. Algebra, lemma etale (cross-chapter, named). Writes the etale R_p-algebra R" as R' tensor_R R_p for an etale R-algebra R'.

Source reference. Original reference 00U2: focused-seeds/geometry/S9-smo-lemma-approximation-property-variant/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original theorem 07QZ: sources/stacks-smoothing/pinned-original-source.tex, source lines 3455–3455. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let R be a Noetherian ring. Let $\mathfrak{p} \subset R$ be a prime ideal. Let $f_1, \ldots, f_m \in R[x_1, \ldots, x_n]$. Suppose that $(a_1, \ldots, a_n) \in ((R_\mathfrak{p})^\wedge)^n$ is a solution. If $R_\mathfrak{p}$ is a G-ring, then for every integer N there exist*

1. *an étale ring map $R \to R'$,*
2. *a prime ideal $\mathfrak{p}' \subset R'$ lying over $\mathfrak{p}$*
3. *a solution $(b_1, \ldots, b_n) \in (R')^n$ in R'*

such that $\kappa(\mathfrak{p}) = \kappa(\mathfrak{p}')$ and $a_i - b_i \in (\mathfrak{p}')^N (R'_{\mathfrak{p}'})^\wedge$.

Proof. By Theorem 07QZ we can find a solution $(b'_1, \ldots, b'_n)$ in some ring R'' étale over $R_\mathfrak{p}$ which comes with a prime ideal $\mathfrak{p}''$ lying over $\mathfrak{p}$ such that $\kappa(\mathfrak{p}) = \kappa(\mathfrak{p}'')$ and $a_i - b'_i \in (\mathfrak{p}'')^N (R''_{\mathfrak{p}''})^\wedge$. We can write $R'' = R' \otimes_R R_\mathfrak{p}$ for some étale R-algebra R' (see Algebra, Lemma 00U2). After replacing R' by a principal localization if necessary we may assume $(b'_1, \ldots, b'_n)$ come from a solution $(b_1, \ldots, b_n)$ in R'. Setting $\mathfrak{p}' = R' \cap \mathfrak{p}''$ we see that $R''_{\mathfrak{p}''} = R'_{\mathfrak{p}'}$ which finishes the proof. □

Date: stacks-project:smoothing.tex:local-snapshot-20261012; GNU FDL 1.2 or later.